

NEW ULM, MN, USA

WMO: 726567

Lat: 44.319N

Lon: 94.502W

Elev: 308

StdP: 97.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04980

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-22.6	-28.9	0.3	-24.7	-27.0	0.3	-22.6	15.1	-10.8	13.7	-9.0	3.6	280	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.3	23.4	30.9	22.8	28.8	21.7	25.5	29.8	24.3	28.5	23.2	27.1	5.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.6	28.2	22.8	18.1	27.1	22.2	17.5	26.3	80.2	29.8	74.9	28.6	70.5	26.5	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.8	9.2	DB	-28.2	35.4	3.3	2.3	-30.5	37.1	-32.4	38.4	-34.3	39.7	-36.7	41.4	(4)
(5)			WB	-28.7	27.6	3.5	1.8	-31.3	28.9	-33.3	29.9	-35.3	30.9	-37.9	32.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-9.6	-7.1	-0.1	7.8	14.6	20.3	22.3	20.7	16.9	9.3	1.2	-6.6
	DBStd	12.52	7.45	7.25	7.19	5.67	4.81	3.68	3.15	3.08	4.67	5.33	6.31	6.80
	HDD10.0	2392	607	479	321	105	13	0	0	0	4	78	271	513
	HDD18.3	4293	865	713	573	317	136	21	5	12	82	284	515	772
	CDD10.0	1499	0	0	7	39	156	309	382	333	209	57	6	0
	CDD18.3	359	0	0	1	2	20	81	128	86	38	4	0	0
	CDH23.3	3228	0	0	3	33	231	717	1161	666	365	51	1	0
	CDH26.7	983	0	0	0	8	73	235	376	174	105	12	0	0
Wind	WSAvg	4.2	4.7	4.7	4.6	4.8	4.5	3.8	3.2	3.0	3.7	4.3	4.5	4.4
Precipitation	PrecAvg	750	18	19	47	70	92	111	106	94	79	53	40	20
	PrecMax	1103	60	70	132	184	216	255	320	264	181	171	124	67
	PrecMin	427	0	1	6	9	18	23	9	26	7	5	1	1
	PrecStd	162	13	15	30	39	47	47	60	46	49	44	31	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	12.3	22.0	27.5	32.2	33.9	34.1	32.6	32.3	28.0	20.9	11.3
		MCWB	3.8	8.3	14.2	15.9	18.5	22.5	25.7	25.0	21.6	17.4	12.7	8.8
	2%	DB	3.0	7.4	17.1	22.8	28.6	31.4	32.3	30.6	29.1	23.9	16.3	7.2
		MCWB	1.3	5.2	12.4	13.2	18.0	21.8	25.1	24.1	20.9	16.0	10.7	4.4
	5%	DB	2.1	4.1	12.8	20.0	26.1	29.0	30.8	28.2	27.5	21.0	12.8	3.7
		MCWB	1.0	2.3	9.4	12.0	16.5	20.6	24.4	22.5	19.8	14.5	8.4	2.0
	10%	DB	0.4	2.5	9.1	17.3	22.9	27.5	28.7	27.2	25.2	17.8	10.0	2.2
		MCWB	-0.3	1.3	6.9	10.4	15.2	19.6	22.8	21.4	18.6	12.9	6.6	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	8.8	16.6	17.1	21.9	25.1	27.9	26.6	24.2	20.7	14.6	9.1
		MCDB	7.4	11.6	18.4	24.5	28.2	30.8	32.1	30.6	28.9	25.0	18.9	10.8
	2%	WB	1.9	5.1	12.7	15.0	19.7	23.7	26.1	24.9	22.8	17.5	11.7	4.3
		MCDB	3.2	7.5	15.8	20.7	24.9	28.6	30.3	28.8	26.9	20.7	15.6	6.4
	5%	WB	0.8	2.9	10.0	13.1	18.1	22.4	24.9	23.7	21.4	15.5	9.1	2.4
		MCDB	1.8	4.1	13.1	18.5	23.0	27.3	29.2	27.3	25.7	19.8	12.5	3.8
	10%	WB	-0.5	1.3	7.1	11.4	16.6	21.1	23.7	22.5	19.9	13.5	7.0	0.9
		MCDB	0.3	2.3	9.4	16.2	21.3	25.6	27.6	25.9	23.9	17.6	9.6	2.0
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.2	8.9	10.7	11.1	10.6	10.6	10.6	12.1	11.1	9.3	8.2
		MCDBR	9.6	10.2	13.8	16.0	15.5	13.3	11.9	11.8	14.2	15.4	13.8	10.7
		MCWBR	8.1	8.1	9.4	8.7	7.3	6.2	6.2	6.4	7.0	8.2	8.9	8.6
	5% WB	MCDBR	8.7	9.4	12.8	13.3	12.2	11.2	10.6	10.6	12.0	12.5	12.5	9.7
		MCWBR	7.5	7.7	9.2	8.2	6.9	6.4	6.2	6.4	7.0	7.8	8.8	8.0
Clear-Sky Solar Irradiance	taub	0.260	0.284	0.306	0.362	0.384	0.401	0.411	0.411	0.370	0.322	0.284	0.264	
	taud	2.472	2.404	2.463	2.344	2.345	2.345	2.305	2.287	2.373	2.509	2.534	2.471	
	Ebn at Noon	885	914	936	901	887	870	855	841	851	859	847	840	
	Edh at Noon	69	88	96	118	122	122	126	124	105	80	65	63	
All-Sky Solar Radiation	RadAvg	1.82	2.85	3.75	4.62	5.39	6.00	6.47	5.35	4.24	2.78	1.80	1.35	
	RadStd	0.20	0.33	0.38	0.46	0.42	0.48	0.24	0.41	0.30	0.36	0.17	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-0.95	-1.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page